

SurgVU Challenge 2026 — Team **Capybara**

Team Report

Quan Huu Cap¹

¹Aillis, Inc., Tokyo, Japan

0.1 Capybara

Code Repository: <https://github.com/huuquan1994/surgvu26-cat2-submission>

Detector-Assisted Tool Verification and Question-Aware Answer Selection for Surgical Visual Question Answering. We extended our winning SurgVU25 solution [1], achieving **0.687276 BERTScore-F1**.

0.1.1 Method Description

We retained our 2025 classifiers and five-frame LLaVA-OneVision-7B pipeline [2]. The VLM generates a description, then answers using it, classifier hints and frames. Inherited components and prompts are documented in our 2025 solution [1].

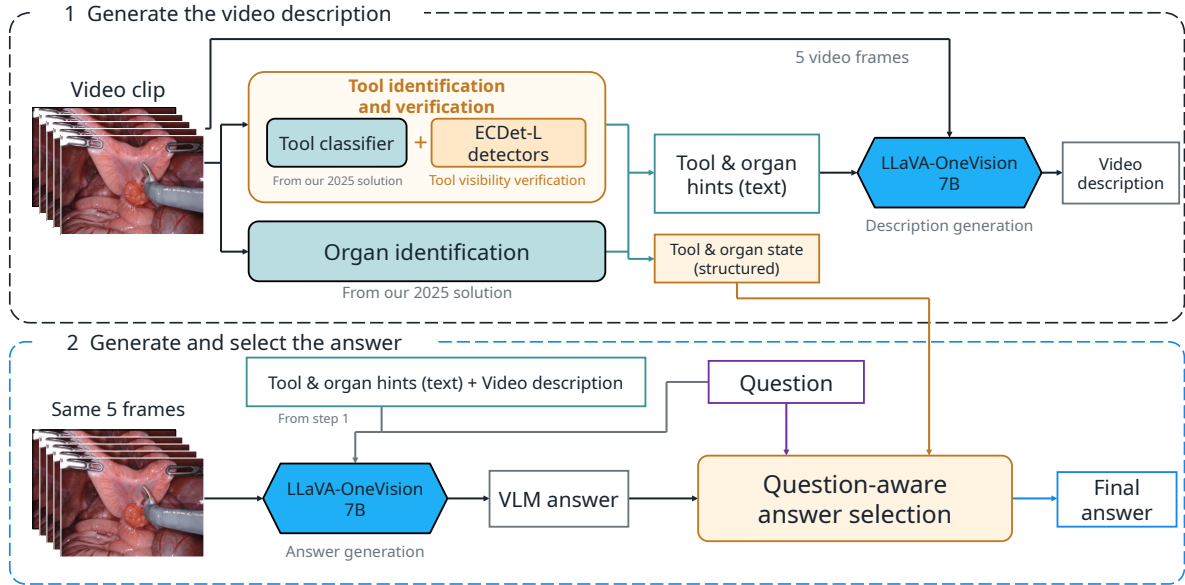


Figure 1: Our SurgVU25 pipeline extended with tool verification and question-aware answer selection, with illustrative frames from our 2025 figure.

Our SurgVU25 tool classifier was trained on installation logs, which record instruments even when they are outside the image. Its false positives can mislead descriptions and answers. To address this, we introduce the EdgeCrafter ECDet-L detectors [3] developed for our SurgVU26 Category-1 entry. We focus verification on cadierre forceps, force bipolar, grasping retractor and prograss forceps because validation showed recurring false positives for these classes. We keep the classifier to retain tool coverage and use ECDet selectively to

reduce false positives. We remove these predictions only when ECDet finds no supporting evidence across the sampled clip.

We retain the 2025 text hints to guide the VLM and add structured evidence for answer selection. The verified tools and organ prediction form *Tool & organ hints (text)*: the tools predicted in each of the five VLM frames, together with the organ label, e.g., “Frame 0: bipolar forceps; organ: kidney.” Both VLM calls receive these hints. *Tool & organ state (structured)* adds tool presence aggregated from a separate 30-frame classifier pass, classifier confidence and the organ label. Answer selection consults these fields directly, avoiding dependence on the VLM expressing them correctly. These are frame-level and clip-level summaries of the same perception pipeline, with separate sampling and aggregation. Converting the state to text cannot recover frame-specific predictions; repeating its tool list for every frame would imply unsupported visibility. We retain frame hints for generation and aggregated state for answer selection, with the five-frame tool list available as a fallback. To prevent the models from reading the displayed tool names, we hide the bottom UI strip before inference.

After both VLM calls, *Question-aware answer selection* uses these outputs according to the question. Presence questions can use the structured tool set, whereas action questions require video interpretation: knowing that forceps are present cannot establish what they are grasping. Table 1 connects this choice to Figure 1.

Table 1: Question-aware answer selection. “State” means Tool & organ state (structured) and “VLM answer” refers to Figure 1. Questions are illustrative.

Type	Example question	How the answer is selected
Presence / list	Is a stapler present?	Read the tools in the state; use fallback information if needed.
Tool identity	What type of forceps is shown?	Combine the VLM answer with tool presence and confidence in the state.
Organ identity	Which organ is shown?	Use the state’s organ label when valid; otherwise use the VLM answer.
Generic purpose	What is a clip applier used for?	Use a predefined explanation for supported general-purpose questions.
Action / relation	What is the forceps grasping?	Use the VLM answer to describe the observed action or relationship.

We keep the visual predictions in generic tool classes and apply commercial naming when constructing answers. A fixed mapping uses the most frequent commercial variant in the tool logs, for example, “bipolar forceps” becomes “maryland bipolar forceps.” For answer formatting, a simple presence question can receive a short statement such as “Yes, a stapler is present.” Tool identities can use just the name when the VLM and state agree. Organ, purpose and action answers retain a short sentence. Invalid outputs fall back to the description or a fixed sentence.

0.1.2 Model Training

We kept the 2025 classifiers and VLM without further training. ECDet-L training used annotated data from our Category-1 entry at 960-pixel resolution. We used data provided by the organizer and public SurgToolLoc-2022 footage; no private external video dataset was used.

0.1.3 Preliminary Performance

We used the official baseline-rescaled **roberta-large** BERTScore-F1 [4], taking the maximum over five references and then averaging over questions. Table 2 reports preliminary and final results separately. Missed detections and incorrect organ or action predictions remain possible. Results do not isolate each addition’s benefit.

Table 2: Challenge performance.

Evaluation	Result (BERTScore-F1)
Prelim phase: rebuilt 2025 pipeline	0.7435
Prelim phase: best preliminary system	0.8663
Final phase: proposed system	0.68728

References

- [1] Quan Huu Cap. Effective surgical visual question answering without ground-truth video descriptions. SurgVU25 Category-2 solution report and code, 2025. <https://github.com/huuquan1994/surgvu25-cat2-submission>.
- [2] Bo Li, Yuanhan Zhang, Dong Guo, et al. LLaVA-OneVision: Easy visual task transfer. *arXiv:2408.03326*, 2024.
- [3] Longfei Liu, Yongjie Hou, Yang Li, Qirui Wang, Youyang Sha, Yongjun Yu, Yinzhi Wang, Peizhe Ru, Xuanlong Yu, and Xi Shen. EdgeCrafter: Compact ViTs for edge dense prediction via task-specialized distillation. *arXiv preprint arXiv:2603.18739*, 2026.
- [4] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*, 2020.